

RESPONSE TO THE REFEREES: NEURAL BELLMAN OPERATORS, R29

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1. OVERVIEW

We thank the referee for identifying both the absent R27 submission object and the scientific attribution problem. This revision supplies a materialized article, supplement, complete computational ledger, historical annex, code, frozen evidence, and executable validation. It retains the title, original economy, original .01 whole-domain target, and every earlier source and result. We respond through proofs, explicit oracles, independently evaluated controls, and complete disclosure of outcomes. We do not treat a response entry, successful compilation, or an auxiliary-model certificate as proof that the original all-state target has been achieved.

The R27 report is the latest report addressed. The R26 findings and the independent R25 second pass are cross-referenced individually below. The original R27 raw-run artifacts were not present at the reviewed commit; the later R28 execution is not represented as recovery of that missing record. The R29 extensions are exploratory relative to all prior outcomes, with fixed cohorts and programmatically enforced reference isolation.

2. R27

2.1. R27-F1 — Materialized referee object

The revision now contains a complete article, essential supplement, computational appendix, response, and historical annex, with source, frozen results, tests, and reproduction instructions. The previous design-only snapshot is not described as a completed manuscript. The current index identifies R29 and the exact reports addressed.

Evidence: Article; SUPP_R29; COMPUTATION_R29; HISTORY_R29; REVISION_MANIFEST.json

Disposition: Correction implemented with the specified evidence.

2.2. R27-F2 — Missing original raw R27 artifacts

The absent R27 raw-run files are not silently reconstructed or assigned retrospective hashes. The existing R28 run is identified by its source and evidence commits. All 34 of its raw policy files are preserved, and the new 81 candidates have their own training records, hashes, dependency audit, and freeze events before reference generation. This supplies a new auditable record, not proof of the missing historical chronology.

Evidence: PROTOCOL.md; PROTOCOL_AMENDMENT_01.md; per-model FREEZE_MANIFEST.json and events.json; R28 frozen inventory records

Disposition: New evidence supplied; missing historical R27 record not retroactively established.

2.3. R27-F3 — Executable exact-head validation

The new validation workflow runs on every push to the new revision branch without a commit-message gate. It checks the frozen data and source identities from a clean checkout, rebuilds every current PDF, and retains the checked SHA and run identity in an external artifact. The separate expensive scientific-generation workflow is not confused with final-head validation.

Evidence: test_revision.py; source_audit/PRESERVATION.json; exact-commit validation artifact

Disposition: Correction implemented with the specified evidence.

2.4. R27-F4 — Generic repair versus neural contribution

Both repair rules are applied to neural, polynomial, ten-seed random-feasible, myopic, base-stock, and two-period tabular proposals. Raw regret, raw residual bounds, correction counts, suffix deficits, topology, and full stage costs are reported separately. The theorem is representation independent. Any neural benefit must be visible in the paired raw quality or burden comparison, rather than inferred from a completed certificate.

Evidence: Article: residual envelopes and state-dependent budgets; controls.py; results/controls; SUPP_R29: attribution table

Disposition: Correction implemented with the specified evidence.

2.5. R27-F5 — Post-diagnostic status

The R27 completion amendment remains identified as post-diagnostic. R28 and R29 are later executions, and the new state-dependent budget rule and expanded controls are exploratory relative to prior outcomes. A committed cohort and programmatic reference isolation are useful reproducibility controls, but they are not described as investigator blinding or retroactive preregistration.

Evidence: PROTOCOL.md; PROTOCOL_AMENDMENT_01.md; per-model FREEZE_MANIFEST.json and events.json

Disposition: Correction implemented with the specified evidence.

2.6. R27-F6 — Small unknown-solution benchmark

The unchanged original four models are retained and the fixed cohort adds dimensions five, six, and seven, reaching 16,384 states per slice. The study measures both approximation and exact-enumeration costs. These larger finite graphs are a scaling diagnostic, not evidence that neural approximation is necessary or that continuous high-dimensional control has been solved.

Evidence: Article: scaling and robustness; SUPP_R29: tolerance panel; COMPUTATION_R29: all candidates

Disposition: The stated extension is established; the broader original scientific requirement remains unestablished.

2.7. R27-F7 — Tabular certification complexity

The paper states feasible and dense state-action-successor counts, $O(\text{TNAD})$ arithmetic enumeration, exact-integer bit growth, and model/policy storage. Exact dynamic programming remains visible in the main scaling table. A complete action-table audit is not claimed to evade the curse of dimensionality.

Evidence: Article: residual envelopes and state-dependent budgets; controls.py; results/controls; COMPUTATION_R29: cost and resource ledgers; results/controls; controls_cli_resource.txt

Disposition: Correction implemented with the specified evidence.

2.8. R27-F8 — Raw four-dimensional failure

The raw neural policies, including the previously reported loss near 0.36656, are preserved. The complete ledger distinguishes raw loss, raw bound, each completed bound, and edited actions for every policy. Fitting, completion, verification, and exact reference costs are separate. The raw failure is neither removed nor relabeled as neural success after repair.

Evidence: Article: scaling and robustness; SUPP_R29: tolerance panel; COMPUTATION_R29: all candidates; COMPUTATION_R29: cost and resource ledgers; results/controls; controls_cli_resource.txt

Disposition: Correction implemented with the specified evidence.

2.9. R27-F9 — Constrained stationarity

The article states the projected-gradient mapping, a precision-aware finite-work theorem on a closed box, the short-face alternative, and explicit unresolved-oracle semantics. Its assumptions are instantiated in two original-economy slices at four meshes with measured oracle calls, validated derivative bounds, and boundary or interior optima.

Evidence: Article: constrained polling and stopped slice; R28 results/slice_summary.json

Disposition: Correction implemented with the specified evidence.

2.10. R27-F10 — Scalar slice versus original policy class

The fixed consumption, zero risky investment, constant preference-adjustment interval, initial state, and two cost coefficients are stated explicitly. The class regret certificate is not substituted for the financed 47-dimensional class or for all states and restarts of the original economy. The economic comparative-static content is retained without changing the original target.

Evidence: Article: constrained polling and stopped slice; R28 results/slice_summary.json; Article: original economic target and conclusion; R25 results/full_state/summary.json

Disposition: Correction implemented with the specified evidence.

2.11. R27-F11 — Stopped derivative bridge

A change-of-path-measure argument gives value, first-derivative, and second-derivative stopping-error bounds for the scalar class, including the explicit parameter dependence of the quadratic cost. Analytic Gaussian-tail and polynomial remainders are enclosed with directed arithmetic. The general financed 47-dimensional derivative-error constants are still not instantiated; finite secants are not presented as that bridge.

Evidence: Article: constrained polling and stopped slice; R28 results/slice_summary.json; Article: financing-gradient formula

Disposition: The stated extension is established; the broader original scientific requirement remains unestablished.

2.12. R27-F12 — Historical-transport isolation

The six executed paths independently implement direct neural differentiation and quotient-gradient pullback with the same carrier, Adam state, and full nonlinear recentering. Every nonlinear remainder is retained and there is no unmatched displacement cap. Parameter, output, moment, and displacement diagnostics are included. This is a path-equivalence experiment, not a cheaper carrier-free optimizer.

Evidence: Article: recentered historical transport; R28 results/central/mechanism_summary.json

Disposition: Correction implemented with the specified evidence.

2.13. R27-F13 — *Production rollback*

The fixed production-schedule study contains twelve gated and ungated trajectories, matched by seed and optimizer, including direct L-BFGS-B. Every checkpoint interval, strict comparison, rejection/restoration record, gradient count, and component timing is retained. The paired final-payoff interval determines whether an improvement is actually proved; restoration correctness alone is not called solver superiority.

Evidence: Article: synchronous deployment; R28 results/central/production_summary.json

Disposition: Correction implemented with the specified evidence.

2.14. R27-F14 — *Original whole-domain target*

The original current-state, all-state, all-restart tolerance remains 0.01. The inherited bound is 7.181834580823298. The revision develops constructive finite-state and scalar stopped-control guarantees but does not claim that they close this original quantitative requirement. Neither the title nor the original economy is replaced by the auxiliary model.

Evidence: Article: original economic target and conclusion; R25 results/full_state/summary.json

Disposition: Original requirement retained and explicitly not claimed as established.

2.15. R27-F15 — *Stale review lineage*

The index now names R29, the R27 report and exact review head, and the R26 and R25 second-pass crosswalks. The previous index is archived byte-for-byte. The missing review files on the R28 base are carried into the new branch as their original Git blobs rather than rewritten summaries.

Evidence: REVISION_INDEX.md; disposition.json; archived referee reports

Disposition: Correction implemented with the specified evidence.

2.16. R27-F16 — *Literature and novelty*

The positioning distinguishes fitted-value-iteration error analysis, constrained generating-set search, verified arithmetic, information relaxation, neural stochastic control, and computational economics. The additions are identified as an explicit residual-budget preservation operator, an interval-oracle boundary theorem with stated work, a stopped scalar oracle, and independently executed mechanism controls. The paper does not claim to invent Bellman comparison, policy iteration, direct search, or interval arithmetic.

Evidence: Article: literature positioning; paper/references.tex

Disposition: Correction implemented with the specified evidence.

2.17. R27-F17 — *Robustness beyond two seeds*

The width-32, depth-two panel has ten seeds at the four-dimensional L=4 model and five at L=5; the original architecture controls remain. Ten random policies per original model, larger dimensions, and a paired three-tolerance panel are added under fixed rules. This is structured computational coverage, not a complete robustness theorem over all charts, optimizers, or economies.

Evidence: Article: scaling and robustness; SUPP_R29: tolerance panel; COMPUTATION_R29: all candidates

Disposition: The stated extension is established; the broader original scientific requirement remains unestablished.

2.18. R27-F18 — *Certified deployment object*

The deployed finite-state object is the exact compiled action table after any replacements. The raw neural network remains the proposal object. An unchanged neural execution certificate would need separate floating-point execution analysis, which is not asserted. Each raw and completed policy has its own hash and edit record.

Evidence: Article: residual envelopes and state-dependent budgets; controls.py; results/controls; per-policy action tables

Disposition: Correction implemented with the specified evidence.

2.19. R27-T1 — *Residual identity and performance bound*

The article defines the policy value, nonnegative local Bellman deficit, and state-sensitive residual envelope. A backward proof covers terminal payoff, settlement, feasible actions, time dependence, and every restart. The new budget theorem additionally proves pointwise improvement over the original candidate and nonnegative remaining allowances.

Evidence: Article: residual envelopes and state-dependent budgets; controls.py; results/controls

Disposition: Correction implemented with the specified evidence.

2.20. R27-T2 — *Information available to completion*

Completion uses the full transition law, exact rewards, exact completed-suffix values, all feasible actions, and exact arithmetic. All of these are stated as powerful model resources also available to exact dynamic programming. Absence of an optimal-value file is not equated with a weaker computational oracle.

Evidence: Article: residual envelopes and state-dependent budgets; controls.py; results/controls; main scaling table

Disposition: Correction implemented with the specified evidence.

2.21. R27-T3 — *Trivial-policy controls*

Random-feasible, myopic, base-stock, and two-period tabular policies are executed under both identical completion rules. The results retain all seeds and failures and report correction burden and costs alongside neural and polynomial proposals.

Evidence: Article: scaling and robustness; SUPP_R29: tolerance panel; COMPUTATION_R29: all candidates

Disposition: Correction implemented with the specified evidence.

2.22. R27-T4 — *Correction topology*

Every changed action records time, state index, original and replacement action, exact suffix deficit, and local allowance. Aggregations distinguish total inventory, minimum inventory, and number of empty coordinates. Restart-specific worst losses and near-default edit counts are reported in the computational appendix.

Evidence: Article: residual envelopes and state-dependent budgets; controls.py; results/controls; COMPUTATION_R29: topology

Disposition: Correction implemented with the specified evidence.

2.23. R27-T5 — *Exact arithmetic cost*

The implementation records arbitrary-precision integer and Fraction backends, logical and dense successor counts, value/denominator bit lengths, wall time, CPU time, model arrays, policy bytes, and scoped process high-water memory. It does not call cumulative RSS an isolated per-case peak. Inherited dual setup time is still marked unmeasured.

Evidence: COMPUTATION_R29: cost and resource ledgers; results/controls; controls_cli_resource.txt

Disposition: The stated extension is established; the broader original scientific requirement remains unestablished.

2.24. R27-T6 — *Remaining-horizon tolerance allocation*

The original fixed eta rule is retained unchanged as a control. The new theorem propagates the already completed suffix envelope and allocates the remaining state-dependent budget, with all-state/all-restart accuracy and policy-preservation proofs. Both rules are evaluated at 0.001, 0.01, and 0.05 on the same tolerance cohort.

Evidence: Article: residual envelopes and state-dependent budgets; controls.py; results/controls; SUPP_R29: tolerance panel

Disposition: Correction implemented with the specified evidence.

2.25. R27-T7 — *Separate numerical stages*

The paper separates candidate generation and compilation, raw audit, completion, completed certification, exact reference, and gap comparison. Full model and reference work are charged to non-amortized component sums. Startup and serialization remain visible at cohort level and are not assigned artificial per-method estimates.

Evidence: COMPUTATION_R29: cost and resource ledgers; results/controls; controls_cli_resource.txt

Disposition: Correction implemented with the specified evidence.

2.26. R27-T8 — *Programmatic reference isolation*

The runtime audit rejects optimal-reference reads during candidate generation and completion, and the independent full-horizon solver is disabled in that context. Candidate, variant, and model-freeze hashes precede reference-creation events. Validation checks these events and file identities. This does not establish recovery of the missing R27 chronology.

Evidence: PROTOCOL.md; PROTOCOL_AMENDMENT_01.md; per-model FREEZE_MANIFEST.json and events.json; controls.py; test_revision.py

Disposition: Correction implemented with the specified evidence.

2.27. R27-T9 — *Strongest classical baseline*

Exact scalar backward induction is in the article scaling table, with model construction and solve times retained. Direct L-BFGS-B remains in the continuous production and historical comparisons. Neither is hidden in an appendix to imply a neural advantage.

Evidence: COMPUTATION_R29: cost and resource ledgers; results/controls; controls_cli_resource.txt; Article: synchronous deployment; R28 results/central/production_summary.json

Disposition: Correction implemented with the specified evidence.

2.28. R27-T10 — Finite versus continuous certification

The finite-state theorem enumerates a graph; the original continuous problem requires domain coverage, derivative or residual enclosures, and a stopped-control comparison. The paper states these as separate complexity regimes. The largest finite graph is not claimed to establish continuous-domain verifier scaling.

Evidence: Article: residual envelopes and state-dependent budgets; controls.py; results/controls; Article: original economic target and conclusion; R25 results/full_state/summary.json

Disposition: Correction implemented with the specified evidence.

2.29. R27-N1 — Complete referee copy

The revision now contains a complete article, essential supplement, computational appendix, response, and historical annex, with source, frozen results, tests, and reproduction instructions. The previous design-only snapshot is not described as a completed manuscript. The current index identifies R29 and the exact reports addressed.

Evidence: Article; SUPP_R29; COMPUTATION_R29; HISTORY_R29; REVISION_MANIFEST.json

Disposition: Correction implemented with the specified evidence.

2.30. R27-N2 — Immutable raw evidence

The absent R27 raw-run files are not silently reconstructed or assigned retrospective hashes. The existing R28 run is identified by its source and evidence commits. All 34 of its raw policy files are preserved, and the new 81 candidates have their own training records, hashes, dependency audit, and freeze events before reference generation. This supplies a new auditable record, not proof of the missing historical chronology.

Evidence: PROTOCOL.md; PROTOCOL_AMENDMENT_01.md; per-model FREEZE_MANIFEST.json and events.json; R28 frozen inventory records

Disposition: New evidence supplied; missing historical R27 record not retroactively established.

2.31. R27-N3 — R26 and R25 crosswalk

The index now names R29, the R27 report and exact review head, and the R26 and R25 second-pass crosswalks. The previous index is archived byte-for-byte. The missing review files on the R28 base are carried into the new branch as their original Git blobs rather than rewritten summaries.

Evidence: REVISION_INDEX.md; disposition.json; archived referee reports

Disposition: Correction implemented with the specified evidence.

2.32. R27-N4 — Attribution-proof completion

Both repair rules are applied to neural, polynomial, ten-seed random-feasible, myopic, base-stock, and two-period tabular proposals. Raw regret, raw residual bounds, correction counts, suffix deficits, topology, and full stage costs are reported separately. The theorem is representation independent. Any neural benefit must be visible in the paired raw quality or burden comparison, rather than inferred from a completed certificate.

Evidence: Article: residual envelopes and state-dependent budgets; controls.py; results/controls; SUPP_R29: attribution table

Disposition: Correction implemented with the specified evidence.

2.33. R27-N5 — *Complete completion proof*

The article defines the policy value, nonnegative local Bellman deficit, and state-sensitive residual envelope. A backward proof covers terminal payoff, settlement, feasible actions, time dependence, and every restart. The new budget theorem additionally proves pointwise improvement over the original candidate and nonnegative remaining allowances.

Evidence: Article: residual envelopes and state-dependent budgets; controls.py; results/controls

Disposition: Correction implemented with the specified evidence.

2.34. R27-N6 — *Executed constrained stationarity*

The article states the projected-gradient mapping, a precision-aware finite-work theorem on a closed box, the short-face alternative, and explicit unresolved-oracle semantics. Its assumptions are instantiated in two original-economy slices at four meshes with measured oracle calls, validated derivative bounds, and boundary or interior optima.

Evidence: Article: constrained polling and stopped slice; R28 results/slice_summary.json

Disposition: Correction implemented with the specified evidence.

2.35. R27-N7 — *Executed faithful transport*

The six executed paths independently implement direct neural differentiation and quotient-gradient pullback with the same carrier, Adam state, and full nonlinear recentering. Every nonlinear remainder is retained and there is no unmatched displacement cap. Parameter, output, moment, and displacement diagnostics are included. This is a path-equivalence experiment, not a cheaper carrier-free optimizer.

Evidence: Article: recentered historical transport; R28 results/central/mechanism_summary.json

Disposition: Correction implemented with the specified evidence.

2.36. R27-N8 — *Paired production rollback*

The fixed production-schedule study contains twelve gated and ungated trajectories, matched by seed and optimizer, including direct L-BFGS-B. Every checkpoint interval, strict comparison, rejection/restoration record, gradient count, and component timing is retained. The paired final-payoff interval determines whether an improvement is actually proved; restoration correctness alone is not called solver superiority.

Evidence: Article: synchronous deployment; R28 results/central/production_summary.json

Disposition: Correction implemented with the specified evidence.

2.37. R27-N9 — *Standalone cost ledger*

The paper separates candidate generation and compilation, raw audit, completion, completed certification, exact reference, and gap comparison. Full model and reference work are charged to non-amortized component sums. Startup and serialization remain visible at cohort level and are not assigned artificial per-method estimates. The earlier central dual-construction cost is not recoverable from its timing records and remains unmeasured.

Evidence: COMPUTATION_R29: cost and resource ledgers; results/controls; controls_cli_resource.txt

Disposition: The stated extension is established; the broader original scientific requirement remains unestablished.

2.38. R27-N10 — *Scaling evidence*

The unchanged original four models are retained and the fixed cohort adds dimensions five, six, and seven, reaching 16,384 states per slice. The study measures both approximation and exact-enumeration costs. These larger finite graphs are a scaling diagnostic, not evidence that neural approximation is necessary or that continuous high-dimensional control has been solved.

Evidence: Article: scaling and robustness; SUPP_R29: tolerance panel; COMPUTATION_R29: all candidates

Disposition: The stated extension is established; the broader original scientific requirement remains unestablished.

2.39. R27-N11 — *Primary-source positioning*

The positioning distinguishes fitted-value-iteration error analysis, constrained generating-set search, verified arithmetic, information relaxation, neural stochastic control, and computational economics. The additions are identified as an explicit residual-budget preservation operator, an interval-oracle boundary theorem with stated work, a stopped scalar oracle, and independently executed mechanism controls. The paper does not claim to invent Bellman comparison, policy iteration, direct search, or interval arithmetic.

Evidence: Article: literature positioning; paper/references.tex

Disposition: Correction implemented with the specified evidence.

2.40. R27-N12 — *Current lineage*

The index now names R29, the R27 report and exact review head, and the R26 and R25 second-pass crosswalks. The previous index is archived byte-for-byte. The missing review files on the R28 base are carried into the new branch as their original Git blobs rather than rewritten summaries.

Evidence: REVISION_INDEX.md; disposition.json; archived referee reports

Disposition: Correction implemented with the specified evidence.

2.41. R27-N13 — *Exact-head CI*

The new validation workflow runs on every push to the new revision branch without a commit-message gate. It checks the frozen data and source identities from a clean checkout, rebuilds every current PDF, and retains the checked SHA and run identity in an external artifact. The separate expensive scientific-generation workflow is not confused with final-head validation.

Evidence: test_revision.py; source_audit/PRESERVATION.json; exact-commit validation artifact

Disposition: Correction implemented with the specified evidence.

3. R26

3.1. R26-F1 — *Latest-review mismatch*

The response identifies the actual R27 report as latest and includes a complete item-by-item R26 and R25 second-pass matrix. The original reports are preserved, not replaced by the stale R24 citation. The revision index and manifest now agree.

Evidence: REVISION_INDEX.md; disposition.json; archived referee reports

Disposition: Correction implemented with the specified evidence.

3.2. R26-F2 — Original 718-fold gap

The original 0.01 target is unchanged, and the retained all-domain bound remains 7.181834580823298. The new finite-state and scalar-class proofs are separate results. No table, abstract, or test flag reports the original target as closed.

Evidence: Article: original economic target and conclusion; R25 results/full_state/summary.json

Disposition: Original requirement retained and explicitly not claimed as established.

3.3. R26-F3 — No demonstrated neural advantage

The strongest exact tabular and direct L-BFGS-B comparators remain in the main text. Raw neural success, repair burden, and total costs are reported separately. The neural representation is not credited with a generic exact-model repair outcome, and numerical superiority is not inferred from feasibility.

Evidence: Article: residual envelopes and state-dependent budgets; controls.py; results/controls; COMPUTATION_R29: cost and resource ledgers; results/controls; controls_cli_resource.txt

Disposition: Correction implemented with the specified evidence.

3.4. R26-F4 — Conditional precision complexity

The full work expression includes the interval-oracle factor $W(\kappa h^2)$. The derivative and payoff oracle is instantiated at the four executed mesh levels of the stopped scalar class. The theorem does not assert arbitrary precision of the 47-dimensional checker or a bare epsilon-inverse-square total-work result.

Evidence: Article: constrained polling and stopped slice; R28 results/slice_summary.json

Disposition: The stated extension is established; the broader original scientific requirement remains unestablished.

3.5. R26-F5 — Boundary exclusion

The projected-gradient theorem includes feasible full coordinate steps and geometric short-face cases at closed-box boundaries. Oracle failure is distinct from an infeasible step. Both a binding upper-face optimum and an interior optimum are executed in the original-economy scalar class.

Evidence: Article: constrained polling and stopped slice; R28 results/slice_summary.json

Disposition: Correction implemented with the specified evidence.

3.6. R26-F6 — Stopped gradient

The scalar likelihood-ratio bridge is proved and instantiated with directed analytic remainder budgets, including explicit cost derivatives. The financing-gradient formula retains all approximation and root-shift terms. Its general 47-dimensional constants remain uninstantiated.

Evidence: Article: constrained polling and stopped slice; R28 results/slice_summary.json; Article: financing-gradient formula

Disposition: The stated extension is established; the broader original scientific requirement remains unestablished.

3.7. R26-F7 — Confounded historical transport

The new independent control recenters to the full nonlinear carrier output after each step, retains the complete remainder, and removes the unmatched output cap. Six paths report parameter, quotient, and moment agreement. The old capped/drifted runs remain archived and are not reinterpreted as faithful controls.

Evidence: Article: recentered historical transport; R28 results/central/mechanism_summary.json

Disposition: Correction implemented with the specified evidence.

3.8. R26-F8 — Post-diagnostic stress versus production

The inherited rate-20.48 stress experiment remains historical engineering evidence. The later production study fixes rates and paired gated/ungated rules before that study, retains all checkpoints, and reports interval comparisons of final payoffs. It is not relabeled as the original R25 held-out test.

Evidence: Article: synchronous deployment; R28 results/central/production_summary.json

Disposition: Correction implemented with the specified evidence.

3.9. R26-F9 — Known-solution manufactured example

The manufactured problem is retained as a verifier stress test with its known-witness information stated. The unknown-solution inventory cohort trains only on model primitives and its own approximate suffix values, then freezes policies before a separately generated exact reference.

Evidence: Article: residual envelopes and state-dependent budgets; controls.py; results/controls; PROTOCOL.md; PROTOCOL_AMENDMENT_01.md; per-model FREEZE_MANIFEST.json and events.json

Disposition: Correction implemented with the specified evidence.

3.10. R26-F10 — High-dimensional verification scaling

The new finite graphs reach dimension seven and 16,384 states and record enumeration, memory, and bit growth. This broadens finite-state coverage. It does not supply a high-dimensional continuous Hessian/interval-cover complexity theorem, which remains a distinct requirement.

Evidence: Article: scaling and robustness; SUPP_R29: tolerance panel; COMPUTATION_R29: all candidates

Disposition: The stated extension is established; the broader original scientific requirement remains unestablished.

3.11. R26-F11 — From-scratch cost

Model construction, proposal generation, compilation, completion, certification, and independent reference are separately recorded for the new inventory executions. Non-amortized totals and resource scope are explicit. The old central dual-construction timing is not invented or inferred from later checker times.

Evidence: COMPUTATION_R29: cost and resource ledgers; results/controls; controls_cli_resource.txt

Disposition: The stated extension is established; the broader original scientific requirement remains unestablished.

3.12. R26-F12 — *Parameterization robustness*

The expanded seed and architecture cohorts, trivial controls, state sizes, and tolerance panels are all retained. The central production study also compares neural and direct optimizers under matched gates. This is wider deterministic coverage, not proof of robustness to every parameter chart or training rule.

Evidence: Article: scaling and robustness; SUPP_R29: tolerance panel; COMPUTATION_R29: all candidates; Article: synchronous deployment; R28 results/central/production_summary.json

Disposition: The stated extension is established; the broader original scientific requirement remains unestablished.

3.13. R26-F13 — *Sparse literature*

The article retains the expanded primary bibliography and adds direct-search stationarity comparisons. Each numerical ingredient is attributed to its established literature, and the contribution statement identifies the added oracle, budget rule, coupling result, and application rather than a generic claim of invention.

Evidence: Article: literature positioning; paper/references.tex

Disposition: Correction implemented with the specified evidence.

3.14. R26-F14 — *Bound-improvement decomposition*

The same-cover warm-start and retained-actor bounds, the separate cover refinement, and the all-restart suffix proof remain in the article and historical annex. A tighter bound is not asserted to prove higher original-domain payoff. Only a strict payoff interval comparison or the finite-model preservation theorem supplies that different conclusion.

Evidence: Article: original economic target and conclusion; R25 results/full_state/summary.json; Article: all-restart suffix envelope

Disposition: Correction implemented with the specified evidence.

3.15. R26-T1 — *Full oracle work*

The work theorem displays $W(\kappa h^2)$ and reports actual oracle work at executed precisions. No headline substitutes an iteration count for total verified work.

Evidence: Article: constrained polling and stopped slice; R28 results/slice_summary.json

Disposition: Correction implemented with the specified evidence.

3.16. R26-T2 — *Oracle failure semantics*

A feasible point with unresolved verification cannot complete a stationarity poll. The algorithm and theorem distinguish this event from an infeasible full step at a nearby face.

Evidence: Article: constrained polling and stopped slice; R28 results/slice_summary.json

Disposition: Correction implemented with the specified evidence.

3.17. R26-T3 — *Exact coefficients versus execution*

Rational policy-table certificates and directed coefficient enclosures are distinguished from floating-point network execution. The repaired finite-state table is the deployed object.

Evidence: Article: residual envelopes and state-dependent budgets; controls.py; results/controls

Disposition: Correction implemented with the specified evidence.

3.18. R26-T4 — *Structurally advantaged comparator*

The manufactured reference remains labeled as known-solution and its direct comparator as structurally informed. The independent unknown-solution experiment is separate; neither universal neural inferiority nor superiority is inferred from the manufactured case.

Evidence: Article: residual envelopes and state-dependent budgets; controls.py; results/controls; historical annex

Disposition: Correction implemented with the specified evidence.

3.19. R26-T5 — *Tests versus theorem hypotheses*

Validation checks executable identities, hashes, fixed cohorts, and recorded numerical enclosures. The manuscript explicitly states that passing these checks does not establish hypotheses that have not been analytically or numerically instantiated.

Evidence: test_revision.py; source_audit/PRESERVATION.json; exact-commit validation artifact; Article: constrained polling and stopped slice; R28 results/slice_summary.json

Disposition: Correction implemented with the specified evidence.

3.20. R26-T6 — *Trajectory distributions*

The transport ledger retains every step and reports displacement distribution, nonlinear remainder, moment mismatch, and carrier alignment. The production ledger retains every candidate interval and deployment decision.

Evidence: Article: recentered historical transport; R28 results/central/mechanism_summary.json; Article: synchronous deployment; R28 results/central/production_summary.json

Disposition: Correction implemented with the specified evidence.

3.21. R26-T7 — *Distinct guarantees*

The paper distinguishes monotone deployed payoff, constrained stationarity, finite-state epsilon optimality, and Adam convergence. Only the corresponding proved guarantee is asserted for each operator.

Evidence: Article: residual envelopes and state-dependent budgets; controls.py; results/controls; Article: constrained polling and stopped slice; R28 results/slice_summary.json; Article: synchronous deployment; R28 results/central/production_summary.json

Disposition: Correction implemented with the specified evidence.

3.22. R26-T8 — *Economic content*

The unchanged stopped economy links costly preference adjustment to a binding boundary at low adjustment cost and an interior solution at the higher cost. This is a verified comparative-static result within the stated class; broader optimal-policy implications are not inferred.

Evidence: Article: constrained polling and stopped slice; R28 results/slice_summary.json

Disposition: Correction implemented with the specified evidence.

3.23. R26-N1 — *R25 second-pass disposition*

The response identifies the actual R27 report as latest and includes a complete item-by-item R26 and R25 second-pass matrix. The original reports are preserved, not replaced by the stale R24 citation. The revision index and manifest now agree.

Evidence: REVISION_INDEX.md; disposition.json; archived referee reports

Disposition: Correction implemented with the specified evidence.

3.24. R26-N2 — *Original quantitative target*

The original 0.01 target is unchanged, and the retained all-domain bound remains 7.181834580823298. The new finite-state and scalar-class proofs are separate results. No table, abstract, or test flag reports the original target as closed.

Evidence: Article: original economic target and conclusion; R25 results/full_state/summary.json

Disposition: Original requirement retained and explicitly not claimed as established.

3.25. R26-N3 — *Unknown-solution stochastic control*

The manufactured problem is retained as a verifier stress test with its known-witness information stated. The unknown-solution inventory cohort trains only on model primitives and its own approximate suffix values, then freezes policies before a separately generated exact reference.

Evidence: Article: residual envelopes and state-dependent budgets; controls.py; results/controls; PROTOCOL.md; PROTOCOL_AMENDMENT_01.md; per-model FREEZE_MANIFEST.json and events.json

Disposition: Correction implemented with the specified evidence.

3.26. R26-N4 — *Instantiate meaningful original-economy control*

The full work expression includes the interval-oracle factor $W(\kappa h^2)$. The derivative and payoff oracle is instantiated at the four executed mesh levels of the stopped scalar class. The theorem does not assert arbitrary precision of the 47-dimensional checker or a bare epsilon-inverse-square total-work result.

Evidence: Article: constrained polling and stopped slice; R28 results/slice_summary.json

Disposition: The stated extension is established; the broader original scientific requirement remains unestablished.

3.27. R26-N5 — *Boundary-aware proof*

The projected-gradient theorem includes feasible full coordinate steps and geometric short-face cases at closed-box boundaries. Oracle failure is distinct from an infeasible step. Both a binding upper-face optimum and an interior optimum are executed in the original-economy scalar class.

Evidence: Article: constrained polling and stopped slice; R28 results/slice_summary.json

Disposition: Correction implemented with the specified evidence.

3.28. R26-N6 — *Faithful mechanism isolation*

The new independent control recenters to the full nonlinear carrier output after each step, retains the complete remainder, and removes the unmatched output cap. Six paths report parameter, quotient, and moment agreement. The old capped/drifted runs remain archived and are not reinterpreted as faithful controls.

Evidence: Article: recentered historical transport; R28 results/central/mechanism_summary.json

Disposition: Correction implemented with the specified evidence.

3.29. R26-N7 — *Production rollback regime*

The inherited rate-20.48 stress experiment remains historical engineering evidence. The later production study fixes rates and paired gated/ungated rules before that study, retains all checkpoints, and reports interval comparisons of final payoffs. It is not relabeled as the original R25 held-out test.

Evidence: Article: synchronous deployment; R28 results/central/production_summary.json

Disposition: Correction implemented with the specified evidence.

3.30. R26-N8 — *Dimension and tolerance scaling*

The new finite graphs reach dimension seven and 16,384 states and record enumeration, memory, and bit growth. This broadens finite-state coverage. It does not supply a high-dimensional continuous Hessian/interval-cover complexity theorem, which remains a distinct requirement.

Evidence: Article: scaling and robustness; SUPP_R29: tolerance panel; COMPUTATION_R29: all candidates

Disposition: The stated extension is established; the broader original scientific requirement remains unestablished.

3.31. R26-N9 — *All infrastructure cost*

Model construction, proposal generation, compilation, completion, certification, and independent reference are separately recorded for the new inventory executions. Non-amortized totals and resource scope are explicit. The old central dual-construction timing is not invented or inferred from later checker times.

Evidence: COMPUTATION_R29: cost and resource ledgers; results/controls; controls_cli_resource.txt

Disposition: The stated extension is established; the broader original scientific requirement remains unestablished.

3.32. R26-N10 — *Primary literature*

The article retains the expanded primary bibliography and adds direct-search stationarity comparisons. Each numerical ingredient is attributed to its established literature, and the contribution statement identifies the added oracle, budget rule, coupling result, and application rather than a generic claim of invention.

Evidence: Article: literature positioning; paper/references.tex

Disposition: Correction implemented with the specified evidence.

3.33. R26-N11 — *Bound decomposition*

The same-cover warm-start and retained-actor bounds, the separate cover refinement, and the all-restart suffix proof remain in the article and historical annex. A tighter bound is not asserted to prove higher original-domain payoff. Only a strict payoff interval comparison or the finite-model preservation theorem supplies that different conclusion.

Evidence: Article: original economic target and conclusion; R25 results/full_state/summary.json; Article: all-restart suffix envelope

Disposition: Correction implemented with the specified evidence.

3.34. R26-N12 — Title and claim alignment

The title Neural Bellman Operators is retained, as are the original economic problem and quantitative target. The abstract and theorem statements identify actual neural-proposal, certified-completion, and continuous-control contributions, rather than replacing the project by a weaker problem or asserting an unsupported neural speedup.

Evidence: Article: residual envelopes and state-dependent budgets; controls.py; results/controls; COMPUTATION_R29: cost and resource ledgers; results/controls; controls_cli_resource.txt

Disposition: Correction implemented with the specified evidence.

4. R25-SP

4.1. R25-SP-F1 — Integrated manuscript

The revision now contains a complete article, essential supplement, computational appendix, response, and historical annex, with source, frozen results, tests, and reproduction instructions. The previous design-only snapshot is not described as a completed manuscript. The current index identifies R29 and the exact reports addressed.

Evidence: Article; SUPP_R29; COMPUTATION_R29; HISTORY_R29; REVISION_MANIFEST.json

Disposition: Correction implemented with the specified evidence.

4.2. R25-SP-F2 — Original whole-domain accuracy

The original current-state, all-state, all-restart tolerance remains 0.01. The inherited bound is 7.181834580823298. The revision develops constructive finite-state and scalar stopped-control guarantees but does not claim that they close this original quantitative requirement. Neither the title nor the original economy is replaced by the auxiliary model.

Evidence: Article: original economic target and conclusion; R25 results/full_state/summary.json

Disposition: Original requirement retained and explicitly not claimed as established.

4.3. R25-SP-F3 — Held-out frontier

The strongest exact tabular and direct L-BFGS-B comparators remain in the main text. Raw neural success, repair burden, and total costs are reported separately. The neural representation is not credited with a generic exact-model repair outcome, and numerical superiority is not inferred from feasibility.

Evidence: Article: residual envelopes and state-dependent budgets; controls.py; results/controls; COMPUTATION_R29: cost and resource ledgers; results/controls; controls_cli_resource.txt

Disposition: Correction implemented with the specified evidence.

4.4. R25-SP-F4 — Faithful moving transport

The new independent control recenters to the full nonlinear carrier output after each step, retains the complete remainder, and removes the unmatched output cap. Six paths report parameter, quotient, and moment agreement. The old capped/drifted runs remain archived and are not reinterpreted as faithful controls.

Evidence: Article: recentered historical transport; R28 results/central/mechanism_summary.json

Disposition: Correction implemented with the specified evidence.

4.5. R25-SP-F5 — Cap-driven nominal learning rates

The old nominal-rate tuning is retained as a result for the capped algorithm, not clean identification of uncapped learning-rate effects. The new recentered transport experiment has no unmatched cap and reports the complete effective output-step distribution, nonlinear remainder, and moment discrepancies. It does not retroactively change the earlier tuning outcome.

Evidence: Article: recentered historical transport; R28 results/central/mechanism_summary.json; Article: synchronous deployment; R28 results/central/production_summary.json

Disposition: Correction implemented with the specified evidence.

4.6. R25-SP-F6 — Known-witness stochastic benchmark

The manufactured problem is retained as a verifier stress test with its known-witness information stated. The unknown-solution inventory cohort trains only on model primitives and its own approximate suffix values, then freezes policies before a separately generated exact reference.

Evidence: Article: residual envelopes and state-dependent budgets; controls.py; results/controls; PROTOCOL.md; PROTOCOL_AMENDMENT_01.md; per-model FREEZE_MANIFEST.json and events.json

Disposition: Correction implemented with the specified evidence.

4.7. R25-SP-F7 — Stopped objective gradient

A change-of-path-measure argument gives value, first-derivative, and second-derivative stopping-error bounds for the scalar class, including the explicit parameter dependence of the quadratic cost. Analytic Gaussian-tail and polynomial remainders are enclosed with directed arithmetic. The general financed 47-dimensional derivative-error constants are still not instantiated; finite secants are not presented as that bridge.

Evidence: Article: constrained polling and stopped slice; R28 results/slice_summary.json; Article: financing-gradient formula

Disposition: The stated extension is established; the broader original scientific requirement remains unestablished.

4.8. R25-SP-F8 — Rollback exercised

The fixed production-schedule study contains twelve gated and ungated trajectories, matched by seed and optimizer, including direct L-BFGS-B. Every checkpoint interval, strict comparison, rejection/restoration record, gradient count, and component timing is retained. The paired final-payoff interval determines whether an improvement is actually proved; restoration correctness alone is not called solver superiority.

Evidence: Article: synchronous deployment; R28 results/central/production_summary.json

Disposition: Correction implemented with the specified evidence.

4.9. R25-SP-F9 — Certificate versus payoff improvement

The same-cover warm-start and retained-actor bounds, the separate cover refinement, and the all-restart suffix proof remain in the article and historical annex. A tighter bound is not asserted to prove higher original-domain payoff. Only a strict payoff interval comparison or the finite-model preservation theorem supplies that different conclusion.

Evidence: Article: original economic target and conclusion; R25 results/full_state/summary.json; Article: all-restart suffix envelope

Disposition: Correction implemented with the specified evidence.

4.10. R25-SP-F10 — Dominant verification and full costs

Model construction, proposal generation, compilation, completion, certification, and independent reference are separately recorded for the new inventory executions. Non-amortized totals and resource scope are explicit. The old central dual-construction timing is not invented or inferred from later checker times.

Evidence: COMPUTATION_R29: cost and resource ledgers; results/controls; controls_cli_resource.txt

Disposition: The stated extension is established; the broader original scientific requirement remains unestablished.

4.11. R25-SP-F11 — Robustness coverage

The width-32, depth-two panel has ten seeds at the four-dimensional L=4 model and five at L=5; the original architecture controls remain. Ten random policies per original model, larger dimensions, and a paired three-tolerance panel are added under fixed rules. This is structured computational coverage, not a complete robustness theorem over all charts, optimizers, or economies.

Evidence: Article: scaling and robustness; SUPP_R29: tolerance panel; COMPUTATION_R29: all candidates

Disposition: The stated extension is established; the broader original scientific requirement remains unestablished.

4.12. R25-SP-F12 — Performance and convergence theorem

The article now supplies complete all-state/all-restart finite-state performance theorems, a projected-stationarity theorem with explicit precision-dependent work, and a rigorously instantiated scalar original-economy oracle. These are constructive guarantees. They do not constitute an epsilon-accurate convergence theorem for the original full continuous-state neural iteration.

Evidence: Article: original economic target and conclusion; R25 results/full_state/summary.json

Disposition: Original requirement retained and explicitly not claimed as established.

4.13. R25-SP-T1 — Current revision index

The index now names R29, the R27 report and exact review head, and the R26 and R25 second-pass crosswalks. The previous index is archived byte-for-byte. The missing review files on the R28 base are carried into the new branch as their original Git blobs rather than rewritten summaries.

Evidence: REVISION_INDEX.md; disposition.json; archived referee reports

Disposition: Correction implemented with the specified evidence.

4.14. R25-SP-T2 — Preserve chronology

The R27 completion amendment remains identified as post-diagnostic. R28 and R29 are later executions, and the new state-dependent budget rule and expanded controls are exploratory relative to prior outcomes. A committed cohort and programmatic reference isolation are useful reproducibility controls, but they are not described as investigator blinding or retroactive preregistration.

Evidence: PROTOCOL.md; PROTOCOL_AMENDMENT_01.md; per-model FREEZE_MANIFEST.json and events.json

Disposition: Correction implemented with the specified evidence.

4.15. R25-SP-T3 — *Effective output-step statistics*

The transport ledger retains every step and reports displacement distribution, nonlinear remainder, moment mismatch, and carrier alignment. The production ledger retains every candidate interval and deployment decision.

Evidence: Article: recentered historical transport; R28 results/central/mechanism_summary.json; Article: synchronous deployment; R28 results/central/production_summary.json

Disposition: Correction implemented with the specified evidence.

4.16. R25-SP-T4 — *Carrier recentering*

The six executed paths independently implement direct neural differentiation and quotient-gradient pullback with the same carrier, Adam state, and full nonlinear recentering. Every nonlinear remainder is retained and there is no unmatched displacement cap. Parameter, output, moment, and displacement diagnostics are included. This is a path-equivalence experiment, not a cheaper carrier-free optimizer.

Evidence: Article: recentered historical transport; R28 results/central/mechanism_summary.json

Disposition: Correction implemented with the specified evidence.

4.17. R25-SP-T5 — *Known-structure comparator*

The manufactured reference remains labeled as known-solution and its direct comparator as structurally informed. The independent unknown-solution experiment is separate; neither universal neural inferiority nor superiority is inferred from the manufactured case.

Evidence: Article: residual envelopes and state-dependent budgets; controls.py; results/controls; historical annex

Disposition: Correction implemented with the specified evidence.

4.18. R25-SP-T6 — *All-restart suffix proof*

The same-cover warm-start and retained-actor bounds, the separate cover refinement, and the all-restart suffix proof remain in the article and historical annex. A tighter bound is not asserted to prove higher original-domain payoff. Only a strict payoff interval comparison or the finite-model preservation theorem supplies that different conclusion.

Evidence: Article: original economic target and conclusion; R25 results/full_state/summary.json; Article: all-restart suffix envelope

Disposition: Correction implemented with the specified evidence.

4.19. R25-SP-T7 — *Deployment versus regret-bound gate*

The paper distinguishes monotone deployed payoff, constrained stationarity, finite-state epsilon optimality, and Adam convergence. Only the corresponding proved guarantee is asserted for each operator.

Evidence: Article: residual envelopes and state-dependent budgets; controls.py; results/controls; Article: constrained polling and stopped slice; R28 results/slice_summary.json; Article: synchronous deployment; R28 results/central/production_summary.json

Disposition: Correction implemented with the specified evidence.

4.20. R25-SP-T8 — No pseudo-replication

Seeds, architectures, dimensions, repeated tolerance checks, time slabs, and verification cells are reported as computational coverage. They are not pooled as independent observations of one statistical treatment effect. Paired repairs and gated/ungated paths retain their dependence explicitly.

Evidence: Article: scaling and robustness; SUPP_R29: tolerance panel; COMPUTATION_R29: all candidates

Disposition: Correction implemented with the specified evidence.

4.21. R25-SP-N1 — Materialize and seal referee object

The revision now contains a complete article, essential supplement, computational appendix, response, and historical annex, with source, frozen results, tests, and reproduction instructions. The previous design-only snapshot is not described as a completed manuscript. The current index identifies R29 and the exact reports addressed.

Evidence: Article; SUPP_R29; COMPUTATION_R29; HISTORY_R29; REVISION_MANIFEST.json

Disposition: Correction implemented with the specified evidence.

4.22. R25-SP-N2 — Original orders-of-magnitude progress

The original current-state, all-state, all-restart tolerance remains 0.01. The inherited bound is 7.181834580823298. The revision develops constructive finite-state and scalar stopped-control guarantees but does not claim that they close this original quantitative requirement. Neither the title nor the original economy is replaced by the auxiliary model.

Evidence: Article: original economic target and conclusion; R25 results/full_state/summary.json

Disposition: Original requirement retained and explicitly not claimed as established.

4.23. R25-SP-N3 — Faithful neural geometry

The six executed paths independently implement direct neural differentiation and quotient-gradient pullback with the same carrier, Adam state, and full nonlinear recentering. Every nonlinear remainder is retained and there is no unmatched displacement cap. Parameter, output, moment, and displacement diagnostics are included. This is a path-equivalence experiment, not a cheaper carrier-free optimizer.

Evidence: Article: recentered historical transport; R28 results/central/mechanism_summary.json

Disposition: Correction implemented with the specified evidence.

4.24. R25-SP-N4 — Proposal-gradient bridge

A change-of-path-measure argument gives value, first-derivative, and second-derivative stopping-error bounds for the scalar class, including the explicit parameter dependence of the quadratic cost. Analytic Gaussian-tail and polynomial remainders are enclosed with directed arithmetic. The general financed 47-dimensional derivative-error constants are still not instantiated; finite secants are not presented as that bridge.

Evidence: Article: constrained polling and stopped slice; R28 results/slice_summary.json; Article: financing-gradient formula

Disposition: The stated extension is established; the broader original scientific requirement remains unestablished.

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The manufactured problem is retained as a verifier stress test with its known-witness information stated. The unknown-solution inventory cohort trains only on model primitives and its own approximate suffix values, then freezes policies before a separately generated exact reference.

Evidence: Article: residual envelopes and state-dependent budgets; controls.py; results/controls; PROTOCOL.md; PROTOCOL_AMENDMENT_01.md; per-model FREEZE_MANIFEST.json and events.json

Disposition: Correction implemented with the specified evidence.

4.26. *R25-SP-N6 — Natural production rejection*

The fixed production-schedule study contains twelve gated and ungated trajectories, matched by seed and optimizer, including direct L-BFGS-B. Every checkpoint interval, strict comparison, rejection/restoration record, gradient count, and component timing is retained. The paired final-payoff interval determines whether an improvement is actually proved; restoration correctness alone is not called solver superiority.

Evidence: Article: synchronous deployment; R28 results/central/production_summary.json

Disposition: Correction implemented with the specified evidence.

4.27. *R25-SP-N7 — Broader robustness and scaling*

The width-32, depth-two panel has ten seeds at the four-dimensional $L=4$ model and five at $L=5$; the original architecture controls remain. Ten random policies per original model, larger dimensions, and a paired three-tolerance panel are added under fixed rules. This is structured computational coverage, not a complete robustness theorem over all charts, optimizers, or economies.

Evidence: Article: scaling and robustness; SUPP_R29: tolerance panel; COMPUTATION_R29: all candidates

Disposition: The stated extension is established; the broader original scientific requirement remains unestablished.